

6.5.6.3, “Matter DN Encoding Rules” for more details. For example:

- Node Subject DN doesn’t include Matter specific **matter-node-id** or **matter-fabric-id** attribute.
- Firmware signing Subject DN doesn’t include Matter specific **matter-firmware-signing-id** attribute.
- Intermediate CA Subject DN doesn’t include Matter specific **matter-icac-id** attribute.
- Root CA Subject DN doesn’t include Matter specific **matter-rcac-id** attribute.
- Certificate Subject DN encodes **matter-node-id** and **matter-rcac-id**, which contradict each other.
- Multiple **matter-cat-id** with the same identifier value and different version numbers, or any **matter-cat-id** with a version number of 0.
- CA certificate doesn’t have Basic Constraints Extension **is-ca** field set to 'true'.
- **key-usage** field of the Key Usage Extension has undefined flags.
- Certificate extension that SHALL be marked as **critical** is marked as **non-critical** in the X.509 representation and vice versa.

6.5.15. Examples

6.5.15.1. Example of Operational Root CA Certificate (RCAC)

The same RCAC in X.509 and Matter TLV formats is presented in this section.

RCAC with Corresponding Private Key in an X.509 PEM Format

```
-----BEGIN CERTIFICATE-----
MIIBnTCCAU0gAwIBAgIIWeqmMpR/VBwwCgYIKoZIzj0EAwIwIjEgMB4GCi sGAQQB
gqJ8AQQMEENBQ0FDQUNBMDAwMDAwMDEwHhcNMjAxMDE1MTQyMzQzWhcNNDExMDE1
MTQyMzQyWjAiMSAwHgYKKwYBBAGConwBBAwQQ0FDQUNBQ0EwMDAwMDAwMTBZMBMG
ByqGSM49AgEGCCqGSM49AwEHA0IABBNT07PvHacIxJCASAF0QH1ZkM4ivE6zPppa
yyWoVgPrptzYITZmpORPWsoT63Z/r6fc3dwzQR+CowtUPdHSS6ijYzBhMA8GA1Ud
EwEB/wQFMAMBAf8wDgYDVROPAQH/BAQDAgEGMBOGA1UdDgQWBBQTr4GrNzdLLtKp
ZJsSt60kKH4VHTAfBgNVHSMEGDAWgBQTr4GrNzdLLtKpZJsSt60kKH4VHTAKBgq
hkjOPQQDAgNIADBFAiBFgWRGbI8ZWrwKu3xstaJ6g/QdN/jV0+7FIKvSoNoFCQIh
ALinwLwELjDPZNww/jNOEgAZZk5RUEkTT1eBI4RE/HUx
-----END CERTIFICATE-----

-----BEGIN EC PRIVATE KEY-----
MHcCAQEEIH1zW+/pFqHAYg4ypib5CZjqqaucQzsom+JnAQdXQaoAoGCCqGSM49
AwEHoUQDQgAEE10js+8dpwjEkIBIAU5AfVmQziK8TrM+mLrLJahWA+um3NghNmak
5E9ayhPrdn+vp9zd3DNBH4KjC1Q90dJLqA==
-----END EC PRIVATE KEY-----
```